

Imperial College London

BSc/MSci EXAMINATION May 2022

*This paper is also taken for the relevant Examination for the Associateship*

## PRINCIPLES OF INSTRUMENTATION

**For 3rd/4th-Year Physics Students**

Monday, 30th May 2022: 14:00 to 16:00

*The paper consists of **three** questions – please answer **all three**.*

*A table of Laplace transforms is included at the end of the paper.*

*Marks shown on this paper are indicative of those the Examiners anticipate assigning.*

### General Instructions

At the top of each page of your answers, write your CID number, module code, question number and page number. Scan and upload your answers to the Turnitin dropboxes as described in the guidance documents in the Blackboard module for this exam. Upload each answer to the dropbox provided for that specific question.

For each answer you should prepare a coversheet which should be the first page of your scanned answer. The coversheet should contain the following:

- your CID
- module name and code
- the question number
- the number of pages in your answer

**You should not write your name anywhere on your answers.**

**You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.**

1. A photomultiplier tube (PMT) set up for photon counting generates a voltage  $v(t)$  across a load resistor  $R_L = 50 \, \Omega$  connected between its anode and ground.

- (i) The PMT response to a single photoelectron is approximately Gaussian with FWHM = 10 ns and peak amplitude of 5 mV. Calculate the PMT gain.

[8 marks]

**ANSWER:** [SEEN] The PMT is a good current source, so the load resistor will not load the anode. The gain is the total charge at the anode generated by a single photoelectron emitted by the photocathode [2]:

$$Q = \int i(t) dt = \int \frac{v(t)}{R_L} dt = \frac{5 \text{ mV}}{50 \, \Omega} \int e^{-\frac{t^2}{2\sigma^2}} dt. \quad [3]$$

Using the Gaussian integral (given) and  $\sigma = \text{FWHM}/2.35 = 4.26 \text{ ns}$ , we have

$$Q = \frac{\sqrt{2\pi} 4.3 \text{ ns } 5 \text{ mV}}{50 \, \Omega} = 1.07 \text{ pC}. \quad [2].$$

Dividing by the electron charge we obtain  $G = 6.66 \times 10^6$  [1].

- (ii) The PMT is now connected to a fast amplifier by a 20 m long coaxial cable with characteristic impedance  $Z_0 = 50 \, \Omega$  and capacitance per unit length  $C' = 100 \text{ pF/m}$ .

- (a) Define 'characteristic impedance' for a generic transmission line and indicate how this is calculated for a lossless coaxial cable. [5 marks]

**ANSWER:** [SEEN] Characteristic impedance of a transmission line is the ratio between the amplitudes of the voltage and resulting current propagating in the line [2]. For a coaxial cable  $Z_0 = \sqrt{L'/C'}$ , where  $L'$  is the inductance per unit length and  $C'$  is the capacitance per unit length of the cable [3].

- (b) Calculate the signal delay in this 20 m long cable. [5 marks]

**ANSWER:** [SEEN] From the above we calculate  $L' = Z_0^2 C' = 2.5 \times 10^{-7} \text{ H/m}$  [2]. Now we can calculate the propagation speed of the cable:  $v = 1/\sqrt{LC} = 2 \times 10^8 \text{ m/s} = 0.67c$ , and hence the delay introduced by 20 m of cable is  $T = 20 \text{ m}/2 \times 10^8 \text{ m/s} = 100 \text{ ns}$  [3].

- (c) The amplifier has input impedance  $Z_i = 75 \, \Omega$ ; characterise the reflected signal (amplitude and polarity) arising from this impedance mismatch for single photoelectron signals. How does this change the single photoelectron response? [5 marks]

**ANSWER:** [SIMILAR TO SEEN] The voltage reflection coefficient at the

[This question continues on the next page ...]

amplifier end is

$$k_r = \frac{Z_2 - Z_1}{Z_2 + Z_1} = 0.2, \quad [1]$$

hence we have a positive (non-inverted) reflection with amplitude of 1 mV [1]. The reflected signal is not actually a problem since at the other end the cable is matched and none of it comes back to the amplifier [2], so the consequence of the mismatch is a slightly larger voltage signal at the amplifier due to the higher input impedance [1].

- (iii) The amplifier has a voltage gain of 20 dB and a constant noise spectral density  $v_n = 1.5 \text{ nV}/\sqrt{\text{Hz}}$  (referred to the input) over its 300 MHz bandwidth (−3 dB).

- (a) Calculate the amplitude of the single photoelectron signal at the output. [5 marks]

**ANSWER:** [SIMILAR TO SEEN] Here we must remember that the equivalent load is now  $Z = 50 \times 75 / (50 + 75) = 30 \Omega$ , so the single photoelectron amplitude is only  $5 \text{ mV} \times 30 \Omega / 50 \Omega = 3.0 \text{ mV}$  [2]. Since the gain is  $G = 10$  (from  $20 \log_{10} 10 = 20 \text{ dB}$ ), the amplitude of the single photoelectron response is 30 mV [3].

- (b) What is the signal-to-noise ratio (in dB) for the single photoelectron peak amplitude? [5 marks]

**ANSWER:** [EXTENSION OF SEEN] Referring to the input, the *rms* noise integrated over the bandwidth is  $v_n = 1.5 \times 10^{-9} \times \sqrt{3 \times 10^8} = 26 \mu\text{V}$  [2], so the S/N ratio is  $30 / 0.026 = 115 = 41 \text{ dB}$  [3].

- (c) Choose the sampling speed and resolution of a suitable digitiser for this system, considering a full scale of 0.1 V. Briefly justify your answer. [7 marks]

**ANSWER:** [EXTENSION OF SEEN] The fast Gaussian response in the time domain implies strictly an infinite frequency spectrum [1]. The amplifier has a bandwidth of 300 MHz, so we can make this our Nyquist frequency – which implies that proper sampling requires at least 600 MS/s [2].

[Note that a calculation based time-bandwidth product is not needed, but give extra marks if students state something sensible here. Considering a time-bandwidth product of 0.44 for a Gaussian we obtain a FWHM of 44 MHz in the frequency domain; note that the correct calculation would involve  $TB=0.88$  for amplitude (0.44 applies to power), but here we want only the single-sided value in the frequency domain hence 0.44 is appropriate.] The S/N ratio is quite high and assuming we want the LSB to be similar to the *rms* noise at the input to the digitiser, which is  $10 \times 26 \mu\text{V} = 0.26 \text{ mV}$  [2]. With 8-bit resolution the LSB corresponds to 0.78 mV, which is quite large,

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while 10-bit resolution would be more appropriate with the LSB at 0.20 mV [2].

[Total 40 marks]

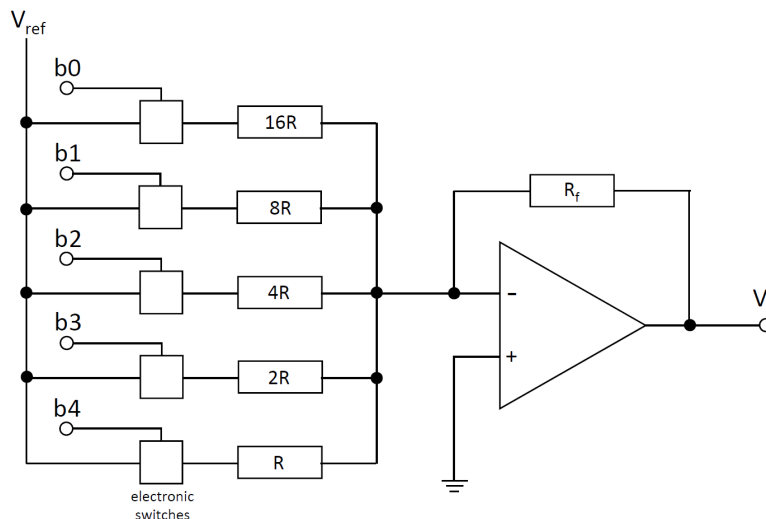
Useful information.

Electron charge  $e = 1.602 \times 10^{-19}$  C.

Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{t^2}{2\sigma^2}} dt = \sqrt{2\pi}\sigma$$

2. Consider the 5-bit Digital-to-Analogue Converter (DAC) shown in the figure, which employs an operational amplifier, initially considered as an ideal device. The DAC converts a binary number  $b_4 b_3 b_2 b_1 b_0$ , where each bit is a binary value (0 or 1), into an analogue voltage  $V_o$ . The bits correspond to voltages of 0 V for a logical '0' and  $V_{ref} = -1$  V for a logical '1'.



- (i) Briefly state the properties of the ideal operational amplifier *and* give the two 'golden-rules' used to simplify the analysis of circuits containing them.

[5 marks]

**ANSWER:** [SEEN] For an op-amp connected with negative feedback [1], i) the potential difference between the inputs is zero [1] (infinite gain [1]) and ii) the inputs draw no current [1] (infinite input impedance [1]).

- (ii) Show that the DAC output can be written as a function of the binary input as:

$$V_o = -V_{ref} \frac{R_f}{R} \left( b_4 + \frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right).$$

[6 marks]

**ANSWER:** [SIMILAR TO SEEN] Since the summing junction is a virtual ground [1] the 5 individual currents are (from top to bottom)  $i_0 = b_0 V_{ref}/16R$ ,  $i_1 = b_1 V_{ref}/8R$ ,  $i_2 = b_2 V_{ref}/4R$ ,  $i_3 = b_3 V_{ref}/2R$  and  $i_4 = b_4 V_{ref}/R$  [2]. Each of these will develop a voltage over the feedback resistor equal to  $-i_n R_f$  [1], and therefore

$$\begin{aligned} V_o &= -R_f(i_0 + i_1 + i_2 + i_3 + i_4) \\ &= -V_{ref} \frac{R_f}{R} (i_0 + i_1 + i_2 + i_3 + i_4) \\ &= -V_{ref} \frac{R_f}{R} \left( b_4 + \frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right). \quad [2] \end{aligned}$$

[This question continues on the next page ...]

- (iii) Derive the value of the feedback resistor  $R_f$  for the DAC to cover a maximum output range of 5 V. Then indicate the voltage corresponding to the least significant bit and the dynamic range of the DAC in dB. [5 marks]

**ANSWER:** [SIMILAR TO SEEN] The maximum output will correspond to b11111 (d31) at the input [1]. using the above equation we can easily calculate  $R_f$ :

$$5 \text{ V} = +1 \text{ V} \frac{R_f}{R} (1 + 1/2 + 1/4 + 1/8 + 1/16) = +1 \text{ V} \frac{R_f}{R} \frac{31}{16},$$

and we obtain  $R_f = 5 R / (31/16) = 2.58 R$  [2]. The lsb is  $5 \text{ V} / 31 = 161.3 \text{ mV}$  [1] and the dynamic range is the full scale divided by the lsb, so  $2^5 = 32$ , or 30.1 dB [1].

- (iv) In some op-amp circuits a resistor ( $R_1$ ) is added between the non-inverting input and ground in order to compensate for the finite impedance of the inputs. Explain how this would work and calculate the resistor value for this design. [6 marks]

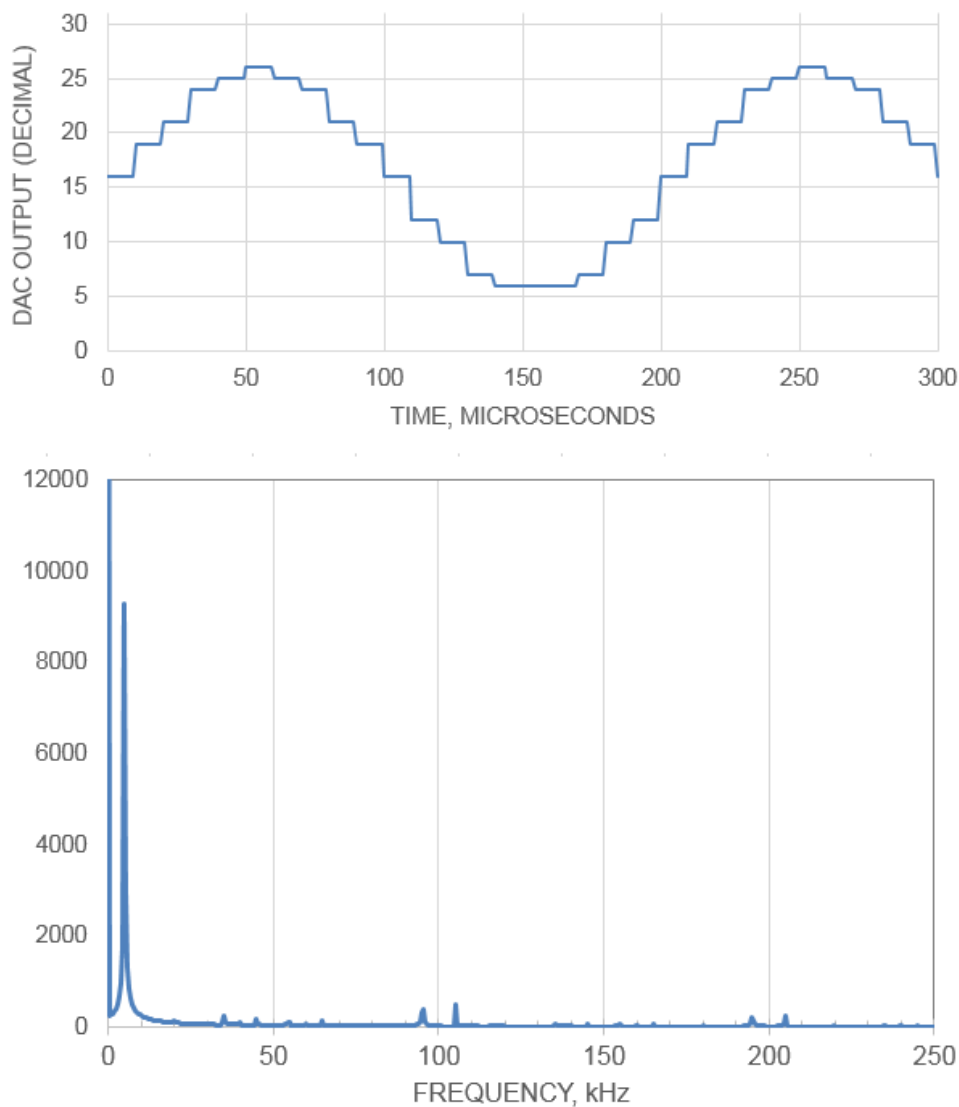
**ANSWER:** [UNSEEN] If there is current entering/leaving the op-amp inputs due to the finite impedance then this will develop a voltage at  $V_-$  or  $V_+$ , and since the op-amp will amplify this difference we would like these voltages to be similar [2]. So, the resistance added to the non-inverting input should match the equivalent resistance seen by the inverting input – and that is the parallel between  $R_f$  and the equivalent resistance to the left of the summing junction, say  $R_{DAC}$  [1]. The equivalent resistance to the left of the junction can be calculated simply if we consider that it is the parallel of 31 resistors ( $1 + 2 + 4 + 8 + 16 = 31$ ) each equal to  $16 R$ , so  $R_{DAC} = 16 R / 31 = 0.52 R$  [1]. Hence:

$$R_1 = R_{DAC} // R_f = \frac{0.52 R \times 2.58 R}{0.52 R + 2.58 R} = 0.42 R. \quad [2]$$

- (v) Consider that a sine wave with period  $200 \mu\text{s}$  is digitised into a 5-bit binary number and fed into the DAC input; the clock time per sample is  $10 \mu\text{s}$ , during which time the input does not change. The amplitude is such that the wave uses most of the full scale available to the DAC. Sketch one period of the DAC output in one plot and the frequency spectrum of the DAC output in another. Explain the most relevant features in each plot, using numerical values where possible. [8 marks]

**ANSWER:** [UNSEEN] The time plot should extent to  $200 \mu\text{s}$  (a little more in the figure below). The vertical axis should show amplitude in some form [1]: a voltage (positive or negative) or a number (binary or decimal); it should have 20 equal time steps [1] and occupy most of the range (to 5 V or d31) [1]. Note: the

[This question continues on the next page ...]



size of these steps is not the lsb if the wave is relatively fast, as is the case here.

The frequency spectrum should contain a DC component at 0 Hz [1], the fundamental frequency of the sine wave at  $1/200 \mu\text{s} = 5 \text{ kHz}$  [1], as well as higher frequencies near  $1/10 \mu\text{s} = 100 \text{ kHz}$  due to the sampling frequency [1] and its harmonics near 200 kHz, 300 kHz, etc., since the digitisation has created sharp edges [1]. There is also aliasing and other more subtle effects which we can ignore.

[Total 30 marks]

3. (i) Give the integral definition of Laplace transform; then derive its time delay property: if a signal  $g(t)$  is equal to another signal  $f(t)$  delayed by time  $T$ , i.e.

$$g(t) = \begin{cases} 0 & 0 \leq t < T \\ f(t - T) & t \geq T \end{cases}$$

show that its Laplace transform is

$$G(s) = e^{-sT} F(s).$$

[5 marks]

**ANSWER:** [SEEN] The integral Laplace transform of a function  $g(t)$  is given by

$$G(s) = \int_0^{\infty} g(t) e^{-st} dt. \quad [2]$$

Therefore,

$$\begin{aligned} G(s) &= \int_T^{\infty} f(t - T) e^{-st} dt \\ &= \int_0^{\infty} f(\tau) e^{-s(\tau+T)} d\tau \\ &= e^{-sT} F(s). \quad [3] \end{aligned}$$

- (ii) Sketch the Bode plot for a lossless transmission line with time delay  $T$ .

[5 marks]

**ANSWER:** [UNSEEN] Starting with

$$X(s) = e^{-sT} Y(s)$$

For the Bode plot we replace  $s \rightarrow j\omega$  [1] and obtain the magnitude and phase:

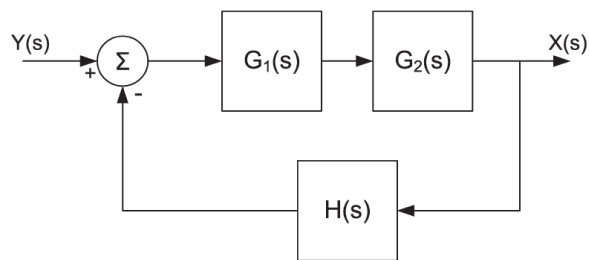
$$\begin{aligned} |e^{-j\omega T}| &= 1 \\ \arg(e^{-j\omega T}) &= -\omega T. \end{aligned}$$

So, the magnitude plot is flat at 0 dB [2] and the phase decreases linearly (becomes more negative) with frequency [2].

- (iii) Consider the negative feedback system represented in the figure below. Show that the system has the following transfer function:

$$T(s) = \frac{G_1(s)G_2(s)}{1 + H(s)G_1(s)G_2(s)}$$

[This question continues on the next page ...]



[5 marks]

**ANSWER:** [SIMILAR TO SEEN] Taking the output of the summing junction to be  $Y'$ , we have [2]

$$\begin{aligned} Y' &= Y - XH \\ X &= Y' G_1 G_2, \end{aligned}$$

from which we obtain

$$\frac{X}{G_1 G_2} = Y - XH.$$

Hence, the transfer function is

$$T = \frac{X}{Y} = \frac{G_1 G_2}{1 + H G_1 G_2}. \quad [3]$$

(iv) Evaluate the transfer function for the following case:

$$\begin{aligned} h(t) &= \delta(t) \\ g_1(t) &= e^{-2t} \\ g_2(t) &= u(t) \end{aligned}$$

[5 marks]

**ANSWER:** [EXTENSION OF SEEN] The three Laplace transforms are [3]

$$\begin{aligned} H(s) &= \mathcal{L}\{\delta(t)\} = 1 \\ G_1(s) &= \mathcal{L}\{e^{-2t}\} = \frac{1}{s+2} \\ G_2(s) &= \mathcal{L}\{u(t)\} = \frac{1}{s} \end{aligned}$$

Hence

$$T(s) = \frac{1}{s(s+2)} \frac{1}{1 + \frac{1}{s(s+2)}} = \frac{1}{s^2 + 2s + 1} = \frac{1}{(s+1)^2}. \quad [2]$$

- (v) For a unit step  $u(t)$  applied to the input at  $t = 0$ , find an expression for the output  $x(t)$  as a function of time for  $t > 0$ . [5 marks]

**ANSWER:** [SIMILAR TO SEEN] For an input  $y(t) = u(t)$  we have  $Y(s) = 1/s$ , and therefore:

$$T(s) = \frac{X(s)}{Y(s)} \therefore X(s) = T(s)Y(s) = \frac{1}{s(s+1)^2} \quad [1]$$

Simplifying by partial fractions,

$$X(s) = \frac{1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$sX(s) \Big|_{s=0} \text{ yields } A = 1;$$

$$(s+1)^2 X(s) \Big|_{s=-1} \text{ yields } C = -1;$$

Substituting in these values

$$X(s) \Big|_{s=1} \text{ yields } B = -1;$$

Hence

$$X(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} \quad [3]$$

Using the given inverse transform,

$$x(t) = u(t) - e^{-t} - t e^{-t}.$$

Since  $t > 0$ ,

$$x(t) = 1 - e^{-t}(1 + t). \quad [1]$$

- (vi) Considering the similarity between the transfer function and the frequency response of a system, calculate how this system would respond to a sinusoidal signal with angular frequency  $\omega$ . [5 marks]

**ANSWER:** [SIMILAR TO SEEN] To obtain the frequency response we simply replace  $s = j\omega$  in  $T(s)$  [2]:

$$T(j\omega) = \frac{1}{(j\omega + 1)^2},$$

and take the magnitude (complex modulus):

$$|T(j\omega)| = \sqrt{TT^*} = \sqrt{\frac{1}{(j\omega + 1)^2} \frac{1}{(-j\omega + 1)^2}} = \frac{1}{\omega^2 + 1}. \quad [3]$$

So, the response at frequency  $\omega$  would be attenuated as per this Lorentzian function.

[Total 30 marks]

Useful information:

In addition to the Laplace transforms at the end of this paper you may need the following inverse transform:

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+a)^n} \right\} = \frac{t^{n-1} e^{-at}}{(n-1)!}$$

| $f(t)$                             | $F(s) = \mathcal{L}\{f(t)\}$        |
|------------------------------------|-------------------------------------|
| $\delta(t)$                        | 1                                   |
| $u(t)$                             | $\frac{1}{s}$                       |
| $t^n$ for $n > 0$                  | $\frac{n!}{s^{(n+1)}}$              |
| $e^{-at}$                          | $\frac{1}{s+a}$                     |
| $te^{-at}$                         | $\frac{1}{(s+a)^2}$                 |
| $\frac{1}{b-a}(e^{-at} - e^{-bt})$ | $\frac{1}{(s+a)(s+b)}$              |
| $\sin(\omega t)$                   | $\frac{\omega}{s^2 + \omega^2}$     |
| $\cos(\omega t)$                   | $\frac{s}{s^2 + \omega^2}$          |
| $e^{-at} \sin(\omega t)$           | $\frac{\omega}{(s+a)^2 + \omega^2}$ |
| $e^{-at} \cos(\omega t)$           | $\frac{s+a}{(s+a)^2 + \omega^2}$    |

Figure 1: Laplace transforms.